Akoizo Original Practice - Form 3 - Answer key

Division B | Solar System | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) A body orbits the Sun at 9 AU. What is its period?

A. 27 years

B. 9 years

C. 81 years

D. 3 years

Correct: A

2. (2 points) Which mission returned a sample of asteroid Bennu?

Unit: Small bodies and the building blocks of life.
Worked-case data: Earth's oceans have a D/H ratio of about 1.6 x 10□^4. Comet 67P's water has about 5.3 x 10□^4, and carbon-rich asteroids have values close to Earth's. Which source better matches Earth's water?
Separate investigation observations:
Sample minerals: Clay minerals and sodium carbonate salts.
Organics: Amino acids and nucleobases detected in the lab.
Isotopes: D/H ratio close to Earth's oceans.

A. OSIRIS-REx

B. Rosetta

C. Dragonfly

D. Kepler

Correct: A

3. (2 points) If a star's surface temperature doubles, its emitted power per area changes by:

A. 4 times

B. 8 times

C. 16 times

D. 2 times

Correct: C

4. (2 points) Silica nanoparticles in Enceladus's plume suggest:

Unit: Ocean worlds: Europa, Ganymede, Enceladus, and Titan.
Worked-case data: A spacecraft flying by a moon detects a magnetic field that changes direction as the moon moves through its planet's field. What does this suggest about the moon's interior?
Separate investigation observations:
Atmosphere: Thick nitrogen, 1.5 times Earth's surface pressure.
Surface temperature: About -180 °C.
Radar: Smooth, dark lakes near the poles.
Probe images: Rounded pebbles on the landing site.

A. Methane lakes

B. Recent asteroid impacts

C. Hydrothermal activity at the seafloor

D. A thick atmosphere

Correct: C

5. (2 points) Hydrogen in a plume along with carbon dioxide is evidence for:

Unit: What makes a world habitable?.
Worked-case data: Europa orbits Jupiter far outside the Sun's habitable zone, yet scientists think it has a liquid ocean. Explain how, using the habitability conditions.
Separate investigation observations:
Location: The moon orbits Saturn, far outside the habitable zone.
Plume: Jets of water vapor and ice grains erupt from cracks.
Plume chemistry: Salts, silica grains, organic molecules, and hydrogen gas.
Orbit: An elliptical orbit kept by resonance with another moon.

A. Chemical disequilibrium as an energy source

B. A thick atmosphere

C. A magnetic field

D. Surface lakes

Correct: A

6. (2 points) Which mission measures gas escaping from Mars's upper atmosphere?

A. MAVEN

B. Perseverance

C. DAVINCI

D. Venus Express

Correct: A

7. (2 points) Why are habitable-zone planets around red dwarfs likely tidally locked?

A. They orbit very close to their dim stars

B. They are very far from their stars

C. Red dwarfs have no gravity

D. They have thick rings

Correct: A

8. (4 points) Explain Kepler's second law in terms of orbital speed.

Unit: Gravity, orbits, and tides.
Worked-case data: Planet A has 4 times Earth's mass and 2 times Earth's radius. Compare its surface gravity and escape velocity with Earth's.
Separate investigation observations:
Orbit: Moon B orbits its planet in 3.6 days on a slightly elliptical path.
Rotation: It always shows the same face to the planet.
Neighbors: Its period is exactly twice that of an inner moon.
Surface: Young, crack-covered ice with few craters.

answer: Equal areas in equal times means the planet must move faster when closer to the star, near periapsis, and slower when farther away. (4 points).

9. (4 points) Why are samples returned to Earth more valuable than remote measurements?

answer: Laboratory instruments on Earth are far more sensitive and can repeat or extend analyses, such as measuring isotopes or detecting amino acids, beyond what a spacecraft can carry. (4 points).

10. (4 points) Explain how albedo affects a planet's equilibrium temperature.

Unit: Light, radiation, and temperature.
Worked-case data: Planet X orbits a Sun-like star at 4 AU with albedo 0.3, the same as Earth. Estimate its equilibrium temperature compared with Earth's 255 K.
Separate investigation observations:
Distance: A planet orbits a Sun-like star at 0.72 AU.
Albedo: Very high: about 0.75, from bright clouds.
Predicted Teq: About 230 K, below freezing.
Measured surface: About 735 K.

answer: Higher albedo reflects more light, so less energy is absorbed and the equilibrium temperature is lower. (4 points).

11. (4 points) Explain why Enceladus's plume is a convenient way to study its ocean.

answer: The plume carries ocean material into space, so a spacecraft can sample it by flying through, without drilling through the ice. (4 points).

12. (4 points) How do salts help liquid water exist in cold places?

Unit: What makes a world habitable?.
Worked-case data: Europa orbits Jupiter far outside the Sun's habitable zone, yet scientists think it has a liquid ocean. Explain how, using the habitability conditions.
Separate investigation observations:
Location: The moon orbits Saturn, far outside the habitable zone.
Plume: Jets of water vapor and ice grains erupt from cracks.
Plume chemistry: Salts, silica grains, organic molecules, and hydrogen gas.
Orbit: An elliptical orbit kept by resonance with another moon.

answer: Dissolved salts lower the freezing point, so brines can remain liquid below 0 °C. (4 points).

13. (4 points) Compare the goals of DAVINCI and VERITAS.

Unit: Venus and Mars: rocky neighbors.
Worked-case data: A Mars orbiter detects clay minerals in layered rocks at the end of a channel that opens into a crater, forming a fan shape. What can you conclude?
Separate investigation observations:
Orbital image: A fan of layered sediment where a valley enters a crater.
Spectra: Clay minerals and carbonates detected in the fan.
Rover data: Fine-grained mudstone layers on the crater floor.

answer: DAVINCI is a probe that descends through the atmosphere to measure its composition and image the surface below. VERITAS is an orbiter that maps topography with radar and identifies rock types globally. (4 points).

14. (4 points) What does the TRAPPIST-1 system show about habitability around red dwarfs?

answer: Red dwarfs can host several rocky planets in the habitable zone, but flares and close orbits may strip atmospheres, as JWST data suggest for the inner planets. (4 points).

15. (4 points) Describe what tidal locking means for climate on an exoplanet.

Unit: Gravity, orbits, and tides.
Worked-case data: Planet A has 4 times Earth's mass and 2 times Earth's radius. Compare its surface gravity and escape velocity with Earth's.
Separate investigation observations:
Orbit: Moon B orbits its planet in 3.6 days on a slightly elliptical path.
Rotation: It always shows the same face to the planet.
Neighbors: Its period is exactly twice that of an inner moon.
Surface: Young, crack-covered ice with few craters.

answer: One side permanently faces the star and the other is in darkness, so heat must be carried by winds or oceans to avoid extreme temperature differences. (4 points).

16. (4 points) Describe one piece of evidence that Bennu's parent body had liquid water.

Unit: Small bodies and the building blocks of life.
Worked-case data: Earth's oceans have a D/H ratio of about 1.6 x 10□^4. Comet 67P's water has about 5.3 x 10□^4, and carbon-rich asteroids have values close to Earth's. Which source better matches Earth's water?
Separate investigation observations:
Sample minerals: Clay minerals and sodium carbonate salts.
Organics: Amino acids and nucleobases detected in the lab.
Isotopes: D/H ratio close to Earth's oceans.

answer: Hydrated clay minerals or evaporite salts formed from brines show water once flowed through the parent body. (4 points).

17. (6 points) Solve this case and explain your reasoning: Planet X orbits a Sun-like star at 4 AU with albedo 0.3, the same as Earth. Estimate its equilibrium temperature compared with Earth's 255 K.

reasoning: Teq is proportional to 1/sqrt d for the same star and albedo.
At 4 AU, sqrt 4 = 2, so Teq is half of Earth's value.
255 K ÷ 2 about 128 K. (3 points).

conclusion: Planet X's equilibrium temperature is about 128 K, far too cold for surface liquid water without another heat source. (3 points).

18. (6 points) Solve this case and explain your reasoning: A spacecraft flying by a moon detects a magnetic field that changes direction as the moon moves through its planet's field. What does this suggest about the moon's interior?

reasoning: A changing planetary field induces currents in an electrical conductor.
Ice and dry rock are poor conductors; salty water is a good conductor.
An induced field responding to the planet's field therefore points to a salty liquid layer. (3 points).

conclusion: The moon probably has a global salty ocean beneath its surface, as Galileo found at Europa. (3 points).